# Classification of States

**Definition. Accessibility**

State $j$ is **accessible** from state $i$, denoted by $i \to j$, if

$$p_{ij}^{(n)} > 0 \quad \text{for some } n \geq 0.$$

**Remark.**

- Accessibility is not a one-step concept: $p_{ij}$ may be zero while $p_{ij}^{(n)} > 0$ for some $n > 1$.
- It is not symmetric: $i \to j$ does not imply $j \to i$.
- Since $p_{ii}^{(0)} = 1$, every state is accessible from itself.

**Definition. Communication**

States $i$ and $j$ **communicate**, denoted by $i \leftrightarrow j$, if

$$i \to j \qquad \text{and} \qquad j \to i.$$

**Note.** Communication is an **equivalence** relation.

**Definition. Communication Class**

A **communication class** is a maximal set $\mathscr{C} \subseteq S$ whose states communicate with one another.

**Note.** Communication classes form a partition of the state space $S$.

**Definition. Irreducible chains**

A Markov chain is **irreducible** if it has exactly one communication class. Otherwise, it is **reducible**.

**Definition. Closed Class**

A communication class $\mathscr{C}$ is **closed** if the chain cannot leave it:

$$p_{ij} = 0 \qquad \text{for every } i \in \mathscr{C} \text{ and } j \notin \mathscr{C}.$$

**Remark.**

- Once $X_n \in \mathscr{C}$ for any $n$, all future states $X_{n+m}$ remain in $\mathscr{C}$ with probability 1, any $m \geq 0$.
- An **absorbing state** $i$ satisfies $p_{ii} = 1$. Hence $\{i\}$ is a closed class.
- A closed class may contain several states and the chain may continue moving inside it.

**Theorem**

Every finite-state Markov chain has at least one closed communicating class.

**Definition. Hitting and First-Return Times**

For a state $j$, define

$$T_j = \inf\{n \geq 0 : X_n = j\}.$$

For a process **starting from state** $i$, define the **first *return* time**

$$T_i^+ = \inf\{n \geq 1 : X_n = i\}.$$

**Remark.**

- $T_j$: the first time to arrive at state $j$.
- $T_i^+$: we know that we start from $i$, and we wonder the first return time to $i$.

**Definition. First-Return Probability**

Starting from $i$, the probability that the first return occurs exactly at time $n$ is denoted as

$$f_{ii}^{(n)} = P_i(T_i^+ = n), \qquad n \geq 1.$$

The probability of ever returning is

$$f_{ii} = P_i(T_i^+ < \infty) = \sum_{n=1}^{\infty} f_{ii}^{(n)}.$$

**Takeaway.** Consider state $i$. If the process returns to $i$ almost surely (with prob. 1), i.e. $f_{ii} = 1$, then it will return to $i$ infinitely many times, and the total returning prob. $\sum_{n=1}^{\infty} p_{ii}^{(n)}$ is infinite. We call $i$ **recurrent**. This is a class property.

**Definition. Recurrence and Transience**

A state $i$ is **recurrent** if

$$P_i(T_i^+ < \infty) = 1.$$

It is **transient** if

$$P_i(T_i^+ < \infty) < 1.$$

Define the number of visits to $i$ after time zero:

$$N_i = \sum_{n=1}^{\infty} \mathbf{1}\{X_n = i\}.$$

**Theorem. Number of Revisits**

For every $m \geq 1$,

$$P_i(N_i \geq m) = (f_{ii})^m.$$

Consequently,

$$\begin{cases} N_i = \infty \text{ a.s.}, & f_{ii} = 1, \\ N_i < \infty \text{ a.s.}, & f_{ii} < 1. \end{cases}$$

$$E_i[N_i] = \begin{cases} \dfrac{f_{ii}}{1 - f_{ii}}, & f_{ii} < 1, \\ \infty, & f_{ii} = 1. \end{cases}$$

Hence,

$$i \text{ recurrent} \iff N_i = \infty \text{ a.s.} \iff E_i[N_i] = \infty,$$

$$i \text{ transient} \iff N_i < \infty \text{ a.s.} \iff E_i[N_i] = \frac{f_{ii}}{1 - f_{ii}}.$$

Define the $n$-step return probability

$$p_{ii}^{(n)} = P_i(X_n = i).$$

**Theorem. Return-Probability Criterion**

$$i \text{ is recurrent} \iff \sum_{n=1}^{\infty} p_{ii}^{(n)} = \infty,$$

$$i \text{ is transient} \iff \sum_{n=1}^{\infty} p_{ii}^{(n)} < \infty.$$

Starting from state $i$,

$$\begin{aligned} i \text{ recurrent} &\iff P_i(T_i^+ < \infty) = f_{ii} = 1 \\ &\iff P_i(N_i = \infty) = 1 \\ &\iff E_i[N_i] = \infty \\ &\iff \sum_{n=1}^{\infty} p_{ii}^{(n)} = \infty. \end{aligned}$$

**Remark.**

- $f_{ii}^{(n)} = P_i(T_i^+ = n)$ counts paths whose *first* return occurs at time $n$.
- $p_{ii}^{(n)} = P_i(X_n = i)$ counts all paths that are at $i$ at time $n$, including paths that returned earlier.

**Note.** So, $f_{ii}^{(n)} \leq p_{ii}^{(n)}$.

**Theorem**

If $i \leftrightarrow j$, then $i$ and $j$ are either both recurrent or both transient.

**Theorem**

For any communication class,

$$\text{Recurrence} \implies \text{closed}.$$

($\Longleftarrow$) holds if the class is also finite.